

EquiBalance User Manual

EquiBalance

Version 1.0 | Draft

Trade Name: EquiBalance

Forage-First Equine Nutrition

[Figure 1: EquiBalance Application Dashboard — Logo/Screenshot Placeholder]

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Author Note

This paper was reviewed using Grammarly to assist with grammar, spelling, and clarity. Anthropic's Claude Sonnet 4.6 was used to assist with drafting and organizing content. All ideas, project details, and final decisions are the original work of the author.

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Purpose of This Guide

EquiBalance is a web application that helps horse owners make nutrition decisions with a forage-first approach, treating hay and pasture as the primary sources of calories rather than commercial feeds. This guide explains how to enter horse, forage, feed, and supplement nutrient information and how to review a nutrition summary with AI-supported explanations. It also provides technical specifications, safety notices, and troubleshooting guidance for both end users and project reviewers.

Safety Notice

⚠ EquiBalance is an educational decision-support tool only. The information provided does not constitute veterinary advice. Your animal's individual needs and circumstances have not been professionally evaluated by this application. Always consult a qualified equine nutritionist, veterinarian, or other licensed animal health professional before making changes to your horse's diet. The developers of EquiBalance make no guarantee of specific health or performance outcomes resulting from use of the application.

Data Privacy

EquiBalance is a cloud app. Horse profile and nutrition data are stored on remote servers protected by SSL/TLS. Data is used only to make your nutrition summary and AI explanation. It is not shared with outside companies for ads. Users must keep their login details safe. For more details, see the EquiBalance Privacy Policy online.

Technical Specifications

Server-Side Infrastructure

EquiBalance runs on a cloud server. The application is set up with these minimum specs:

- Processor: 4 virtual central processing units (vCPUs)
- Memory: 16 GB RAM
- Storage: 100 GB solid-state drive (SSD) cloud storage
- Operating System: Linux (latest stable distribution)
- Network: Standard cloud-provider network interface with outbound internet access

This setup supports user sessions, database operations, nutrition calculations, and AI API integration without resource limits (Mell & Grance, 2011).

Client-Side Requirements

EquiBalance runs entirely in the browser. No software installation is required. The minimum client requirements are:

- Device: Desktop computer, laptop, tablet, or smartphone
- Browser: A current-version modern browser (Google Chrome, Mozilla Firefox, Apple Safari, or Microsoft Edge)
- Internet: Stable broadband or cellular data connection

Software Dependencies

Server-side software needed for EquiBalance includes:

- Web server framework (e.g., Node.js/Express or Python/Django)
- Relational database management system (e.g., PostgreSQL)
- AI/LLM API integration for the interactive chat module
- SSL/TLS certificate for encrypted HTTPS communication

Relation to Other Documents

This manual is part of the project documents for the EquiBalance capstone in ITCC298. Companion documents are referenced here.

Document	Relationship to This Guide
Week 2 — Capstone Proposal (Trader, 2026a)	Defines EquiBalance purpose, intended users, features, and the forage-first feeding philosophy that underlies the application.
Week 3 — Hardware Requirements Report (Trader, 2026b)	Defines cloud-based server specifications, operating system, RAM, storage, and user device requirements. Source for the Technical Specifications section of this manual.
Sample Horse Profile	Shows a completed example profile for reference when completing Step 2.
Sample Forage Analysis	Shows the type of laboratory data users may enter in Step 3.

Sample Nutrition Summary	Shows an example of output generated by the application in Step 5.
EquiBalance Privacy Policy	Full details on data collection, storage, and user rights. Available on the EquiBalance website.

A Software Requirements Specification is planned. It will detail more functional and non-functional requirements.

Before You Begin

What You Need

Before starting, gather the following:

- A desktop, laptop, tablet, or smartphone with a modern web browser
- Reliable internet access
- Horse information: weight, age, workload, and feeding goals
- Forage information: hay type, pasture details, or hay analysis results
- Feed and supplement information: product names and daily feeding amounts

Step 1: Open EquiBalance

- **Open a modern web browser** (Chrome, Firefox, Safari, or Edge).
- **Go to the EquiBalance web address:** www.equibalance.com
- **New user:** Select Create an Account and complete the registration form.
- **Returning user:** Select Sign In and enter your credentials.
- **On the Home screen, select** Start New Profile.

[Figure 2: Home Screen — Placeholder Screenshot]

- Review the Welcome Message and Safety Notice.
- **Select** Continue.

Step 2: Create a Horse Profile

Required fields have asterisks (*). Optional fields improve nutrition summary accuracy.

[Figure 3: Horse Profile Input Form — Placeholder Screenshot]

- **Select** Create Horse Profile.
- Enter the horse's name.*
- Enter the horse's estimated body weight.*
- Select the horse's age category.*
- Select the horse's workload.*
- Enter the horse's body condition score (BCS). (Optional. See Glossary.)
- Add notes about health history, feeding goals, or management concerns.
(Optional)
- **Select** Save Profile.

Step 3: Add Forage Information

[Figure 4: Forage Analysis Entry Screen — Placeholder Screenshot]

- Open the saved horse profile.
- **Select** Add Forage.
- Choose Hay, Pasture, or Other forage.
- Enter the forage source or description.
- Enter the amount fed per day.
- Enter forage analysis values if available. Useful values include crude protein (%), non-structural carbohydrates (NSC, %), digestible energy (Mcal/kg), and mineral levels. These are typically found on a laboratory hay analysis report.
- **If no analysis is available, select** Use Estimated Values.
- **Select** Save Forage.

Step 4: Add Feed or Supplement Information

[Figure 5: Feed and Supplement Database Search Screen — Placeholder Screenshot]

- **Select** Add Feed or Supplement.
- Search for the product name.
- Select the correct feed, ration balancer, or supplement from the results.
- Enter the amount fed per day.
- Confirm the serving unit.

- **Select** Save Feed

Note: If a product does not appear in search results, it may not be in the EquiBalance database. Enter the product manually or select a similar feed category as an alternative.

Step 5: Review Nutrition Summary

[Figure 6: Nutritional Gap Results Dashboard — Placeholder Screenshot]

- Open the horse profile.
- **Select** View Nutrient Summary.
- Review the forage, feed, and supplement totals.
- Check for highlighted nutrient gaps (shown in red).
- Check for highlighted nutrient excesses (yellow highlights).
- Review calories, protein, vitamins, and minerals.
- **To save a copy, select** Save Summary.

Step 6: Read the AI-Supported Explanation

[Figure 7: AI Chat Interface — Placeholder Screenshot]

- **From the nutrient summary screen, select** View Explanation.
- Read the plain-language summary of your horse's nutrition profile.

- Review the explanation of possible gaps or excesses.
- Review the suggested next steps.
- **If you need more information, select** Ask a Follow-Up Question.

Note: AI-generated responses are not a substitute for professional veterinary or nutritional guidance. Always consult a qualified equine professional before making significant changes to your horse's diet.

Editing and Deleting Entries

You can update or remove any saved entry at any time.

To Edit an Entry

Open the horse profile, locate the entry you want to change, and select the pencil icon (). Update the relevant fields and select Save.

To Delete an Entry

Open the horse profile, locate the entry you want to remove, and select the trash can icon (). Confirm the deletion when prompted. This action cannot be undone.

Warning and Status Messages

Message	What It Means	What to Do
"Horse profile is incomplete."	A required field in the horse profile is missing.	Return to the horse profile, complete the missing required fields (marked with *), and save.
"Forage data is incomplete."	The forage entry is missing the amount, type, or analysis values.	Return to the forage entry and add the missing information.
"Estimated values are being used."	The app is using general averages instead of lab analysis results.	For more accurate results, enter values from a laboratory forage analysis report.
"Results may be limited."	The app does not have enough data to produce detailed guidance.	Check that the horse profile is complete, forage amounts have been entered, and at least one feed or supplement has been saved.

Troubleshooting

Problem	Possible Cause	Solution
The nutrition summary does not appear.	Required information is missing.	Complete all required fields in the horse profile and ensure at least one forage entry has been saved.
The results seem too general.	The app uses estimated forage values.	Enter specific forage analysis data or add more detailed feed and supplement information.
A feed or supplement is not listed.	The product may not be in the database.	Enter the product manually or select a similar feed category.
The app loads slowly.	Slow or unstable internet connection.	Check your internet connection and try refreshing the page. If the problem continues, try again later as the server may be temporarily busy.
The AI explanation is unclear.	The profile or forage data may be incomplete.	Add more specific data and select Ask a Follow-Up Question to request a more detailed explanation.

Interface Icons

Icon	Meaning
+ (Plus, sign)	Add a new item
 Pencil	Edit an existing entry
 Trash can	Delete an entry (this action cannot be undone)
 Warning triangle	Missing or incomplete data — action required
 Check mark	Entry saved successfully
 Information	Additional help or explanation available
 Chat bubble	Opens the AI-powered chat assistant for guided nutrition support
 Save / Disk	Saves the current horse profile or feed ration to your account

Glossary

Term	Definition
Body condition score (BCS)	A numerical scale (typically 1–9) used to assess the amount of fat on a horse's body. A score of 5 is considered ideal for most horses.
Crude protein	A measure of the total protein content in a feed or forage, expressed as a percentage of dry matter.
Digestible energy (DE)	The amount of energy a horse can extract from a feed, expressed in megacalories per kilogram (Mcal/kg).

Forage analysis	A laboratory report providing nutrient values for a specific batch of hay or pasture, including protein, energy, minerals, and carbohydrate fractions.
Forage-first nutrition	A feeding philosophy that prioritizes hay and pasture as the main sources of calories and nutrients, with commercial feeds used only to fill specific gaps.
Non-structural carbohydrates (NSC)	A measure of sugar and starch content in forage or feed. Particularly relevant for horses with metabolic conditions such as equine metabolic syndrome or laminitis.
Ration balancer	A concentrated feed product designed to be fed in small amounts to supply vitamins, minerals, and protein without adding significant calories.

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Submission Feedback

Overall Feedback

Dannie, this is one of the most professionally structured drafts in the class. The glossary, the troubleshooting table, the warning and status messages table, and the AI-disclosure author note are all elements I see in shipped enterprise documentation, not student work. Your safety notice was particularly well done - you correctly framed EquiBalance as decision-support, not veterinary advice, and that distinction is exactly what would keep your future product out of an FDA or FTC inquiry. There is a 2021 JAMA study showing that medical decision-support tools with clearly worded "consult a professional" disclaimers see roughly 50 percent better clinician adoption because professionals trust them. Same dynamic applies to equine nutrition. One thing for the final - your figures are still placeholders ("Figure 1: Logo/Screenshot Placeholder"). Replace them with actual mockup images before you submit the final capstone. The forage-first philosophy is a real differentiator and the way you tied each section back to it was strong. -Prof W